

Business Problem Statement

Amazon FBA & FBM Seller Performance Dashboard

Background

The business operates a multi-category Amazon storefront selling 10 SKUs across five categories (Kitchen, Sports, Home Office, Pet Supplies, and Electronics Accessories) through a mix of Fulfillment by Amazon (FBA) and Fulfillment by Merchant (FBM) channels. Over the past 12 months, the store generated **\$579.98K** in gross revenue and **\$285.68K** in net profit (a 49.26% margin), supported by \$12.24K in advertising spend. Despite this apparent profitability on paper, several underlying figures in the dataset raise questions about whether the business is actually as healthy as the top-line numbers suggest, and whether its current growth is sustainable.

Core Business Problem

Leadership cannot currently tell, from revenue and profit alone, **where money is being made, where it is quietly leaking, and whether the customer base being built today will still be profitable a year from now.** A previously identified red flag — customer acquisition cost exceeding lifetime value — has since been corrected after a measurement error was found, but four tensions in the data still warrant attention:

- 1. Customer economics are positive, but thin.** Customer Lifetime Value (\$96.66) now exceeds Customer Acquisition Cost (\$66.52), a 1.45:1 ratio — correcting an earlier measurement error that had shown the relationship inverted. The business is not losing money on acquisition, but 1.45:1 sits well below the 3:1 ratio commonly used as a benchmark for a sustainably scalable acquisition model, leaving little room for CAC to rise before the model turns unprofitable.
- 2. Growth is concentrated, not diversified.** One region (US-West) is not just underperforming but currently **unprofitable** (-45.90% margin on \$5.36K revenue), against \$135K–\$152K and 40%+ margins in the other three regions. A small number of SKUs and categories dominate the revenue mix. It is unclear whether this is a demand issue, a stock availability issue, or a listing/visibility issue.
- 3. Retention is weak relative to acquisition spend.** Repurchase rate sits at 6.98% with roughly 2,051 churn events recorded against 6,000 total customers. With CAC/CLV no longer the most urgent issue, retention is now the clearer lever: lifting repurchase rate is what would move the CLV:CAC ratio from 1.45:1 toward a healthier 3:1.
- 4. Ad spend is not clearly translating to incremental revenue.** Ad-driven revenue accounts for only 0.84% of total revenue despite \$12.24K in spend and 5K attributed orders, while overall TACOS sits at a lean 2.11%. Sponsored Display shows the strongest return (51.16%) despite the smallest budget, suggesting spend may be misallocated across campaign types rather than underperforming overall.

Questions Each Dashboard Section Must Answer

The dashboard is built around four analytical sections. Each one exists to answer a distinct question that rolls up into the core problem above.

1. Profit & Sales

- Is the business actually profitable once COGS, fulfillment fees, ad spend, and refunds are all accounted for — not just on gross revenue?
- Which categories, products, and regions are driving profit, and which are dragging it down?
- Is revenue growth (the Nov–Dec spike) durable, or is it a seasonal effect that will reverse?
- Is the 2.72% refund rate concentrated in specific products or channels, signaling a listing or quality issue?

2. Advertising / PPC

- Is every advertising dollar spent recovering more than it costs, by campaign type and category?
- Why is ad-driven revenue only 0.84% of total revenue — is this a healthy mostly-organic business, or a sign that ads are underperforming and budget should shift toward what already works?
- Which product categories return the most per ad dollar, and should budget be reallocated toward them?
- Are click-through and conversion rates trending in a direction that will require creative or listing changes?

3. Inventory & Operations

- Which SKUs are at risk of stocking out, and which are tying up cash in overstock (0.96 turnover, 5.75% stockout rate)?
- Are FBA fees eating into margin disproportionately for specific categories?
- Are the 139 inbound shipment defects concentrated in a supplier, product, or time period that can be fixed?
- Is the business holding the right amount of inventory to support demand without triggering long-term storage fees?

4. Customers

- Now that CLV exceeds CAC (1.45:1), what would it take to move that ratio toward the 3:1 benchmark — higher CLV through retention, or lower CAC through channel reallocation?
- Which acquisition channel produces customers worth keeping (highest CLV), not just the most volume?
- What is driving the 2,051 churn events, and is churn concentrated in a specific product category or month?
- Is the repurchase rate improving or declining month over month, and is the business in Acquisition Mode or Loyalty Mode as a result?

What Success Looks Like

This dashboard should let a stakeholder move from a top-line number to a root cause in under two clicks. Concretely, it succeeds if it can answer the following at any time:

Metric	Current Value	Question It Must Keep Answering
Net Profit Margin	49.26%	Is margin holding as revenue scales, or compressing?
CAC vs. CLV	\$66.52 vs. \$96.66 (1.45:1)	Is the ratio climbing toward the 3:1 healthy benchmark?
Repurchase Rate	6.98%	Is retention improving month over month?
Stockout Rate	5.75%	Which SKUs are driving it, and is it improving after restock?
TACOS	2.11%	Is ad spend scaling total sales, or just ad-attributed sales?

Refund Rate	2.72%	Is it concentrated in specific categories or regions?
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Prepared as the guiding problem statement for the Amazon FBA & FBM Analytics Dashboard (Profit & Sales, Advertising, Inventory & Operations, Customer Metrics).